ALOK RAM TRIPATHI

FULL-STACK SOFTWARE ENGINEER · AWS CERTIFIED DEVELOPER (ASSOCIATE)

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PROFESSIONAL SUMMARY

Full-stack software engineer and AWS Certified Developer (Associate) with 2+ years of experience building production web and GenAI applications used by 10,000+ people. Own features end to end: SEO-optimized React and Next.js frontends, Python and Node.js REST APIs, PostgreSQL data models and RAG pipelines, deployed as containerized services on AWS with Docker and CI/CD.

PROFESSIONAL EXPERIENCE

Software Engineer · MandS Consulting May 2025 – Present

- Own end-to-end delivery of features for two client products, Braided (5,500+ users) and Hey Coach (4,500+ users, growing monthly), from design review to production release.
- Build a production RAG pipeline for Braided with OpenAI embeddings and pgvector, using semantic chunking and similarity search to ground LLM answers in product data, reduce hallucinations, and stream cited responses.
- Engineer server-rendered, SEO-optimized Next.js and React interfaces that improve search crawlability and page-load times on high-traffic pages.
- Create a shared Tailwind CSS component library that speeds up feature delivery and enforces one design system across both products.
- Optimize PostgreSQL schemas, indexes, and queries behind complex business workflows for faster responses and reliability as usage grows.
- Develop Python REST APIs connecting the frontend, backend, and database layers for 10,000+ combined users.
- Ship a cross-platform Flutter mobile app with a responsive UI and full REST API integration.
- Modernize legacy modules to remove technical debt and speed up page loads, working with product and design to scope and deliver on time.

MERN Stack Developer Intern · Vector India Pvt. Ltd. Jun 2024 – Mar 2025

- Built full-stack apps with MongoDB, Express.js, React.js, and Node.js, implementing JWT authentication, CRUD REST APIs, and client-side routing.
- Deployed to Vercel and Render through Git and GitHub workflows and contributed to peer code reviews.

PROJECTS

Docent: RAG Knowledge Assistant · Next.js, FastAPI, pgvector, OpenAI, Claude on Bedrock github.com/Aloktripathi19/Rag-Assistant

- Built a full-stack RAG app to chat with uploaded PDF, TXT and Markdown files: answers are grounded only in the documents, streamed over SSE, and carry clickable citations to the exact passage and page.
- Designed the ingestion and retrieval pipeline: page-bounded token-aware chunking, batched OpenAI embeddings, HNSW cosine search in pgvector, and query condensing for follow-up questions.
- Added a model picker (OpenAI GPT models or Claude Sonnet 4.5 on AWS Bedrock); containerized the FastAPI backend with Docker and set up GitHub Actions CI for lint, type-checks, tests and builds.

TECHNICAL SKILLS

LANGUAGES	JavaScript, TypeScript, Python, SQL, HTML5, CSS3
FRONTEND	React.js, Next.js, Tailwind CSS, Bootstrap, Flutter, Responsive and SEO-Optimized UI
BACKEND	Node.js, Express.js, FastAPI, REST APIs, JWT Authentication
AI / GENAI	RAG Pipelines, pgvector, Embeddings, Semantic Search, LLM APIs, OpenAI, Claude API, AWS Bedrock
DATABASES	PostgreSQL, Supabase, MySQL, MongoDB
CLOUD, DEVOPS	AWS (ECS, ECR, EC2, VPC, Route 53, ALB, IAM, CodePipeline, CodeBuild), Docker, CI/CD
TOOLS	Git, GitHub, Bitbucket, Jira, Postman, Agile/Scrum, Code Review

CERTIFICATIONS

- AWS Certified Developer – Associate · Amazon Web Services
- Anthropic Academy: Building with the Claude API · Model Context Protocol: Advanced Topics · Claude with Amazon Bedrock
- AI/ML for Geodata Analysis, Participation Certificate · ISRO

EDUCATION

B.Tech, Information Technology · Mahakal Institute of Technology Aug 2019 – May 2023

CGPA 7.50 / 10 Ujjain, Madhya Pradesh